GROVE: Reusing Solution Sketches for Current-Flow Group Selection

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***Abstract*—Current-flow group closeness centrality (CFCC) selects network anchors by minimizing the grounded Laplacian inverse trace. Greedy optimization repeatedly estimates every remaining vertex's marginal contribution, requiring expensive solves after each selection. We present GROVE, which reuses two solution sketches through rank-one grounding updates. Unrefreshed candidates use the current sketches, while refreshed candidates retain inverse moments obtained through individual numerical solves. A per-round cap limits additional candidate refreshes. A uniform concentration argument, combined with residual bounds, yields a $(1-1/e-\varepsilon)$ approximation to optimal gain relative to a fixed seed under explicit sufficient conditions, whose sketch dimension is far above the empirical $d = 64$; $\varepsilon > 0$ is the accuracy parameter. On six graphs with completed comparisons, GROVE is 1.1–5.7× faster than ApproxGreedy in the main runs, and 1.4–3.8× faster in four-seed means on four graphs. Final inverse traces differ by at most 1.6% across the compared methods; the forest methods reach lower traces on roadNet-PA, and the smallest-eigenvalue heuristic of Zhou et al. is faster on DBLP. Runtime components and ablations show where reuse saves work and where updates or refreshes remain costly. On LiveJournal, with 4.0 million vertices, GROVE finishes in 4626 seconds. Supporting structural results cover completely monotone grounding objectives, and inverse-power experiments characterize the tradeoff between aggregate resistance and slow spectral modes.**

***Index Terms*—Group centrality, node grounding, completely monotone functions, supermodular optimization, randomized sketches, incremental computation.**

I. Introduction

Resource placement and leader selection [23]–[25] require groups of vertices whose collective influence depends on paths throughout a network. Current-flow group closeness centrality (CFCC) measures this influence through effective resistance [1]. Let $G = (V, E)$ be a connected undirected graph with $n = |V|$ vertices and Laplacian L. For a nonempty anchor set $S \subseteq V$, let $L_{-S} = L[V \setminus S, V \setminus S]$ be its grounded principal submatrix. Maximizing $\mathrm{CFCC}(S) = n/\operatorname{tr}(L_{-S}^{-1})$ is equivalent to minimizing the total effective resistance to the anchors.

The computational bottleneck lies in choosing each additional anchor. Greedy selection compares how much grounding each remaining vertex decreases the inverse trace, and these marginal gains change after every selection. ApproxGreedy estimates them through random projections and Laplacian solves [1]; forest-based methods use sampled spanning forests [5]. Although each round adds only one vertex, both repeat all-candidate estimation for the updated grounded system. The central question is how to reuse this work without letting stale estimates misdirect later choices.

Two natural approaches expose the difficulty. Rebuilding solution sketches every round estimates the current matrix but repeats expensive batches of solves. Alternatively, a scalar cache estimates each candidate's moments—its inverse diagonal and squared inverse-column norm—once, then subtracts algebraic decrements after each selection. The initial sketch error persists in absolute terms. As true moments shrink, their relative errors grow and may distort the ranking, and the cache gives no way to tell beforehand on which inputs this happens. The shortcut is competitive on most graphs we test and fails on one.

GROVE combines two forms of reuse. It maintains two solution sketches of the current grounded inverse, updating them through rank-one identities rather than rebuilding them through solves. Unrefreshed candidates use these current sketches. Refreshed candidates instead retain moments obtained from individual numerical solves. Unlike the scalar-cache shortcut, this second path starts from a solved column rather than a noisy initial sketch: exact decrements preserve exact moments in exact arithmetic, and residual bounds control the numerical implementation. A per-round refresh cap limits additional candidate solves. The resulting algorithm maintains current estimates while preserving work already spent on accurate candidates.

Reuse creates an adaptive estimation problem: selected vertices depend on the same randomness used in subsequent rounds. We cover all feasible grounding sets in one concentration event and combine it with numerical-error bounds. For a fixed seed S_0, k additional anchors, and accuracy parameter $\varepsilon > 0$, sufficient sketch and residual conditions yield a $(1-1/e-\varepsilon)$ approximation to the optimal gain relative to S_0. These are conservative worst-case conditions; the practical sketch dimension of $d = 64$ is evaluated empirically. Our work bound separately accounts for sketch updates and candidate refreshes.

We evaluate GROVE on seven SNAP networks [15] against ApproxGreedy, our implementations of ForestCFCM and SchurCFCM, and Zhou's smallest-eigenvalue heuristic [6]. Matched comparisons, runtime components, ablations, parameter sweeps, and additional random seeds test selection cost and quality. The results show substantial savings on several graphs, a smaller

advantage on web-Google, and small quality tradeoffs rather than uniform dominance. The largest completed input has about four million vertices.

As a supporting extension, we study completely monotone (CM) trace objectives. Hard-CFCC supermodularity and hardness are already known [1]; our structural result covers all inverse powers and heat traces under grounding. Increasing the inverse exponent emphasizes slower modes. Separate exact-marginal experiments use this parameter to trade aggregate resistance against the smallest grounded eigenvalue. These objective choices complement the CFCC algorithm.

Contributions.

1) **Hybrid CFCC selection.** Maintained solution sketches, retained refreshed moments, and capped refresh reduce repeated marginal-estimation work.
2) **Approximation and work.** A uniform adaptive event and numerical-error budgets give a conditional seeded-gain guarantee, with explicit refresh-sensitive costs.
3) **Evaluation and objective extensions.** Same-machine comparisons and component measurements assess the reuse mechanism; CM grounding structure and integer-power experiments explain the separate spectral-weighting option.

II. Problem and Preliminaries

Table I collects shared notation.

A. Grounding queries

Let $G=(V,E,w)$ be an undirected weighted graph with $n=|V|$ vertices and $m=|E|$ edges. Its nonnegative weights satisfy $w_{ij}=w_{ji}$, with $w_{ij}=0$ for nonedges. The Laplacian has entries $L_{ii}=\sum_j w_{ij}$ and $L_{ij}=-w_{ij}$ for $i\neq j$. Write $M=L+\sigma I$, where I is the identity and $\sigma\geq 0$ is a uniform regularization parameter. For a selected set $S\subseteq V$, let $\mathbf{1}_S$ be its indicator vector and $\operatorname{diag}(\mathbf{1}_S)$ the matrix with this vector on the diagonal. A grounding penalty $\beta\in(0,\infty]$ defines

$$A_S^\beta=\begin{cases} M+\beta\operatorname{diag}(\mathbf{1}_S), & \beta<\infty,\\ M[V\setminus S, V\setminus S], & \beta=\infty. \end{cases} \tag{1}$$

Finite β penalizes the selected vertices (soft grounding); hard grounding removes their rows and columns. Write $X_S=A_S^\infty$ for this active principal submatrix. When $\sigma=0$, assume connectivity and a nonempty fixed seed S_0; when $\sigma>0$, an empty seed is admissible. We consider supersets of S_0, whose nonempty grounded matrices are positive definite. The trace of a function of the empty matrix is defined as zero.

A finite-valued function $f:(0,\infty)\to[0,\infty)$ is *completely monotone* (CM) if it admits the representation

$$f(x)=\int_{[0,\infty)} e^{-tx}\,\mu(dt), \qquad \mu\geq 0, \tag{2}$$

where μ is a nonnegative measure on $[0,\infty)$ and t is the integration variable. The matrix function applies f to each eigenvalue. The algorithmic target is hard CFCC: $\beta=\infty$, $\sigma=0$, and $f(x)=x^{-1}$; the other choices support the structural extension. Starting from S_0, we select k additional anchors:

$$\min_{T\subseteq V\setminus S_0,\ |T|=k} \Phi_f(S_0\cup T), \qquad \Phi_f(S)=\operatorname{tr} f(A_S^\beta). \tag{3}$$

The *seeded gain*, normalized relative to S_0, is $F_f(T)=\Phi_f(S_0)-\Phi_f(S_0\cup T)$; its maximum over feasible k-element extensions is OPT_k. All approximation factors below compare extensions of this same seed. The experimental total budget is $K=|S_0|+k$: one seed and $K=20$ total anchors mean $k=19$ additions.

For a candidate $v\notin S$, its marginal cost reduction is $\Delta_f(v\mid S)=\Phi_f(S)-\Phi_f(S\cup\{v\})$. Supermodularity means $\Delta_f(v\mid S)\geq\Delta_f(v\mid U)$ whenever $S\subseteq U$ and $v\notin U$. Thus the seeded gain is submodular; when it is also monotone, exact greedy selection achieves the standard $1-1/e$ approximation to OPT_k [10].

B. Current flow and spectral weighting

For hard grounding with $\sigma=0$, injecting one unit of current at an ungrounded vertex u produces voltage $(L_{-S}^{-1})_{uu}$ at u, equal to its effective resistance to the grounded set. Summing these resistances gives the denominator of CFCC [1], so minimizing $\Phi_1(S)=\operatorname{tr}L_{-S}^{-1}$ maximizes CFCC.

More generally, inverse powers $f(x)=x^{-s}$, $s>0$, give $\Phi_s(S)=\operatorname{tr}(A_S^\beta)^{-s}$. If λ_i are the positive eigenvalues of A_S^β, then $\Phi_s(S)=\sum_i\lambda_i^{-s}$. For $\lambda_i<\lambda_j$, their relative weights are $(\lambda_j/\lambda_i)^s$; increasing s therefore emphasizes slower spectral modes. Heat instead weights each mode by $e^{-\tau\lambda_i}$ at horizon $\tau>0$. CFCC is our main computational and experimental objective. The broader family provides interpretable spectral weights.

The hard-grounding CFCC cost Φ_1 is nonincreasing and supermodular, and exact group selection is NP-hard [1]. These known properties are all that the greedy algorithm and its analysis in Sections III and IV require; Section V extends them to the CM family.

III. The Hybrid CFCC Algorithm

We now specialize to hard grounding with $\sigma=0$ and CFCC ($s=1$). To avoid rebuilding all marginal estimates after every anchor, GROVE maintains solution sketches for unrefreshed candidates and scalar moments for candidates already refreshed by a solve. The two paths preserve different information about the current matrix (Fig. 1(b)).

Let $W=V\setminus S_0$ contain the $N=n-|S_0|$ initial candidates. The seed matrix is $A_0=L[W,W]=C^\top C\in\mathbb{R}^{N\times N}$, where $C\in\mathbb{R}^{m\times N}$ is the oriented edge-incidence matrix, scaled by square roots of edge weights and restricted to columns in W. This exact incidence factor is distinct from the approximate Cholesky factor used for preconditioning. For additional anchors $T\subseteq W$, let R_T be the current active inverse, embedded in $\mathbb{R}^{N\times N}$ with zero rows and columns at T. Write $R=R_T$, let e_u be the coordinate unit vector at an active candidate u, and define $r_j(u,T)=e_u^\top R_T^j e_u$ for integer $j\geq 1$. The CFCC marginal is

$$\Delta_1(u\mid S_0\cup T)=\frac{r_2(u,T)}{r_1(u,T)}, \qquad r_1=R_{uu}, \quad r_2=\|Re_u\|^2. \tag{4}$$

TABLE I
NOTATION SHARED ACROSS SECTIONS. AUXILIARY PROOF VARIABLES ARE LOCAL TO THEIR STATEMENTS.

Symbol	Meaning	Symbol	Meaning
L, M	Laplacian and $M = L + \sigma I$	s, p	Real inverse exponent and positive integer exponent
S_0, N	Fixed seed and $N = n - \lvert S_0 \rvert$ initial candidates	Φ_s^0	Initial inverse-power cost $\Phi_s(S_0)$
T, k, K	Added anchors, additional budget, total budget	$\Omega(u,v)$	Pairwise effective resistance
A_0, C, R_T	Seed matrix, incidence factor, current padded inverse	Y_1, Y_2	CFCC solution sketches for first/second moments
Π_C, Π_I	Independent random projection matrices	$\varepsilon, \varepsilon_{\text{forest}}$	GROVE accuracy parameter and forest-method input
$e_u, r_j(u,T)$	Coordinate vector and $e_u^\top R_T^j e_u$	$\bar{R}_v, s_x$	Inverse after selecting v, and $\lVert Re_v \rVert^2$
$d, B_{\text{init}}, B_r, b_r$	Sketch rows, initial/refresh batch widths, refresh cap	θ, ζ, δ	Acceptance, moment-error, and failure tolerances
$n_{\text{ref}}, n_t^{\text{ref}}$	Distinct refreshed candidates in total/in round t	$\alpha_{\text{step}}, \alpha_{\text{cost}}$	Greedy-step fraction and trace-cost factor
B_0, B_U	Seed approximate inverse and active restriction	$\mathscr{E}, \rho_{\text{res}}$	Sketch-error matrix and column-residual bound
$F_c, \mathscr{T}_t$	Factor entries and work of a solve in round t	$z_{\text{ser}}, \vartheta$	Formal-series variable and removal interpolation

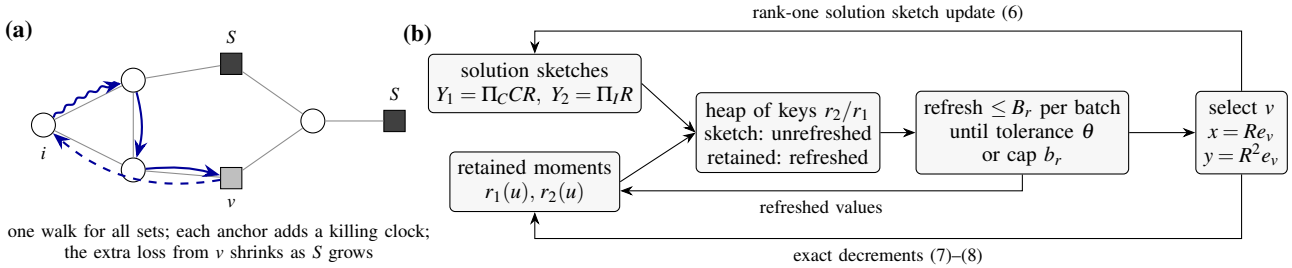


Fig. 1. (a) The CM coupling: all grounding sets share one walk and one family of killing clocks, so a new anchor removes fewer surviving returns when more anchors are present. (b) The maintained state of GROVE: unrefreshed candidates are keyed from the current solution sketches, refreshed candidates from retained exact moments; after each selection both paths are updated from the selected inverse columns.

Here $\lVert \cdot \rVert$ is the Euclidean norm. Both moments come from one inverse column, so a candidate refresh requires one solve rather than two complete trace estimates.

A. Common-seed initialization

The initial anchor approximately minimizes L^+_{vv}, where L^+ is the Moore–Penrose pseudoinverse. Writing $\Omega(u,v)$ for pairwise effective resistance, the identity $\sum_u \Omega(u,v) = \operatorname{tr} L^+ + nL^+_{vv}$ gives the singleton CFCC rule [1], [5]. We estimate this diagonal with an incidence-based random-sign sketch, projecting right-hand sides and solutions onto $\mathbf{1}^\perp = \{x : \mathbf{1}^\top x = 0\}$, where $\mathbf{1}$ is the all-ones vector. Every compared method receives the common seed $S_0 = \{s_0\}$. Its computation is a separate shared stage; all later gains use this seed as their fixed baseline.

Initializing scalar estimates once saves solves, but subtracting exact decrements preserves their initial absolute error. As moments shrink, this error can distort their ratios. GROVE instead maintains sketches of the *current* inverse. Draw independent random matrices $\Pi_C \in \mathbb{R}^{d\times m}$ and $\Pi_I \in \mathbb{R}^{d\times N}$ with independent mean-zero entries of variance $1/d$. Starting at $R = A_0^{-1}$, construct the two $d \times N$ solution sketches

$$Y_1 = \Pi_C CR, \qquad Y_2 = \Pi_I R, \qquad \widehat{r}_j(u) = \lVert Y_j[:,u] \rVert^2 \quad (j = 1,2). \tag{5}$$

The notation $Y_j[:,u]$ denotes column u. The identity $RA_0R = R$ gives $\lVert CRe_u \rVert^2 = r_1(u)$ and $\lVert Re_u \rVert^2 = r_2(u)$, so the squared column norms estimate the two current moments. Initialization uses d sparse-system right-hand sides per solution sketch.

B. Two complementary update paths

Grounding one vertex gives a rank-one inverse correction, which can be applied directly to both solution sketches. An *unrefreshed* candidate uses the updated solution-sketch column norms. A *refreshed* candidate has undergone an individual solve and thereafter retains its moments through scalar updates. The following identities describe both paths in exact arithmetic.

Proposition 1 (Hybrid state identities)**.** *On selecting v, let $x = Re_v$, $y = R^2 e_v$, $c = 1/x_v$, and $s_x = \lVert x \rVert^2$. The next inverse is $R' = R - cxx^\top$, with a zero row and column at v. Both solution sketches update as*

$$Y_j' = Y_j - cY_j[:,v]x^\top, \qquad j = 1,2. \tag{6}$$

For each remaining refreshed candidate u, retain its moments by

$$r_1'(u) = r_1(u) - cx_u^2, \tag{7}$$

$$r_2'(u) = r_2(u) - 2cx_u y_u + c^2 s_x x_u^2. \tag{8}$$

With exact columns, exact refreshed moments remain exact. A scalar cache initialized with error $e_j(u)$ retains that signed absolute error under exact independent decrements. The solution sketch update costs $O(Nd)$ work and storage; scalar retention costs $O(N)$ work, once x, y are available.

Proof sketch. The principal-inverse identity gives $R' = R - cxx^\top$. Multiplication by the fixed sketch factors gives (6). Taking diagonals gives (7); squaring first gives (8). Subtracting the same decrement from a true moment and its cached estimate preserves their signed difference. □

Algorithm 1 GROVE: hard-CFCC hybrid selection

Input: G, common seed s_0, total budget K; $d, B_{\text{init}}, B_r, b_r, \theta$; solve tolerances $\varepsilon_{\text{init}}, \varepsilon_{\text{ref}}, \varepsilon_{\text{upd}}$.

1. Ground s_0; construct the seed preconditioner if enabled. Initialize Y_1, Y_2 and moments using batches of width B_{init} and tolerance $\varepsilon_{\text{init}}$. Mark all candidates unrefreshed.
2. For rounds $1, \dots, K-1$:
   a. Build heap $\mathscr{H}$ with key $r_2(u)/r_1(u)$. Set incumbent value $b = -\infty$, winner $v = \perp$, refresh count $r = 0$.
   b. Consume refreshed heap tops and update (v, b) from their values. Let q_{heap} be the largest remaining key, or $-\infty$ if empty.
   c. If $v \neq \perp$ and $b \geq (1-\theta) q_{\text{heap}}$, stop this round. If $v \neq \perp$, $b_r > 0$, and $r \geq b_r$, also stop.
   d. Pop entries until up to B_r unrefreshed candidates are collected; refreshed entries encountered update the incumbent without a solve. Refresh the collected candidates at ε_{ref}, mark them refreshed, update (v, b), and retain a newly winning column Re_v. Increase r by the batch size and return to step 2b.
   e. Before a nonfinal round ends, obtain $x = Re_v$ and $y = R^2 e_v$. Reuse a retained winning column when enabled; solve any missing columns at ε_{upd}. Update both solution sketches by (6). For unrefreshed candidates recompute moment norms; for refreshed candidates apply (7)–(8).
   f. Zero coordinate v, add v to the grounded set, and mask it in subsequent solves. Skip maintenance after the final selection.
3. Return the K grounded nodes, including s_0.

After each selection, GROVE updates both solution sketches for all remaining vertices and the scalar moments of refreshed candidates. Retention allows a later round to choose an already refreshed candidate without another refresh. A winning refresh provides x; the additional column y supports scalar maintenance. Both are reused when available. Numerical solves introduce errors controlled by the tolerances below and analyzed in Section IV.

C. Capped candidate refresh

At each round, rebuild a max-heap from the current moment ratios, using solution sketch estimates for unrefreshed candidates and retained values for refreshed ones. Refreshed heap tops update the incumbent without a solve; unrefreshed candidates are refreshed in blocks of at most B_r. Let b be the best refreshed marginal found and q_{heap} the largest remaining key. Stop when $b \geq (1-\theta) q_{\text{heap}}$, with acceptance tolerance $0 \leq \theta < 1$, or when the positive refresh cap b_r is reached. A zero cap means unlimited refreshes. The cap controls candidate solves, not solution sketch maintenance. It is checked between batches, so a positive cap can be exceeded by fewer than B_r refreshes; the initial maximum-key candidate is always processed before a cap stop.

Algorithm 1 uses total budget K, initialization batch width B_{init}, and refresh batch width B_r. The relative solve tolerances $\varepsilon_{\text{init}}, \varepsilon_{\text{ref}}, \varepsilon_{\text{upd}}$ apply to initialization, candidate refreshes, and newly computed update columns, respectively. A reused winning column retains its ε_{ref} tolerance. The seed preconditioner is constructed once and masked for later solves.

Classical lazy greedy (CELF) reuses stale exact marginal upper bounds from submodularity [11]. A raw JL estimate is not such an upper bound: combining ApproxGreedy with lazy ordering requires calibrated intervals or fresh accuracy events. GROVE instead updates current estimates and retains individually refreshed moments; its cap is analyzed on the uniform event in Section IV. Two restrictions isolate the update paths in Section VI. *Sketch-only* maintains both solution sketches but rekeys every candidate from them each round, removing the retained scalar path and its y solve. *Scalar cache* initializes moments once and applies algebraic decrements without maintaining solution sketches. Their comparison tests whether current sketches and retained values reduce different sources of repeated work.

IV. Approximation and Work

Reuse saves solves but couples later choices to the original random sketches. We first control this adaptivity, then incorporate numerical errors, summarize the solver policy, and account for the work that remains after eliminating repeated all-candidate solves.

A. One event for adaptive selection

Accuracy for a fixed matrix is insufficient when the selected anchors depend on the sketches. We therefore cover every feasible grounding set in one event. Let δ be its failure probability and ζ the relative moment-error tolerance. The count $\mathscr{N}_{N,k}$ below includes every active candidate paired with a set of at most k added anchors; α_{step} is the resulting per-round greedy factor.

Theorem 2 (Adaptive hybrid guarantee)**.** *Fix S_0, $1 \leq k < N$ additions, $0 < \delta, \zeta < 1$, and define*

$$\mathscr{N}_{N,k} = \sum_{t=0}^{k} \binom{N}{t} (N-t), \qquad d \geq \left\lceil 8\zeta^{-2} \log \frac{4\mathscr{N}_{N,k}}{\delta} \right\rceil.$$

Use independent Gaussian entries of variance $1/d$, or independent signs $\pm 1/\sqrt{d}$, in the two solution sketches. With probability at least $1-\delta$, exact maintenance gives relative error at most ζ for both moments at every active vertex of every $|T| \leq k$. For exact refreshed values, put $\chi = (1+\zeta)/(1-\zeta)$ and

$$\alpha_{\text{step}} = \min\{\chi^{-2}, (1-\theta)\chi^{-1}\}.$$

Algorithm 1, with any positive cap or unlimited refreshes, returns gain at least $(1-e^{-\alpha_{\text{step}}})\text{OPT}_k$. In particular, $\zeta \leq \varepsilon/8$ and $\theta \leq \varepsilon/2$ give $(1-1/e-\varepsilon)\text{OPT}_k$. The sufficient dimension is $O(\varepsilon^{-2}((k+1)\log(N+1) + \log(1/\delta)))$.

Proof sketch. One fixed column. For a deterministic nonzero target vector w and one unscaled random sketch row g, the normalized projection $Z = g^\top w/\|w\|$ satisfies $\mathbb{E}Z^2 = 1$ and $\mathbb{E}e^{tZ} \leq e^{t^2/2}$ for both row distributions, hence $\mathbb{E}e^{tZ^2} \leq (1-2t)^{-1/2}$ for $0 < t < 1/2$. A Chernoff bound on the average of d copies of Z^2 gives upper-tail failure at most $e^{-d\zeta^2/8}$; the inequality $e^{-y} \leq 1 - y + y^2/2$ and $\mathbb{E}Z^4 \leq 3$ give lower-tail failure at most $e^{-d\zeta^2/6}$.

All adaptive trajectories. The targets $CR_T e_u$ and $R_T e_u$, over all $|T| \leq k$ and active u, form a *deterministic* family of $2\mathscr{N}_{N,k}$ vectors. A union bound puts all of them inside relative error ζ with probability at least $1-\delta$. The maintained solution sketches equal $\Pi_C C R_T$ and $\Pi_I R_T$ exactly for whatever T the

algorithm reaches, so adaptive selection, refresh patterns, and stopping decisions stay on this event.

One round. Write Δ_u for candidate u's true current marginal. Every key lies in $[\Delta_u/\chi, \chi\Delta_u]$, and refreshed keys are exact. Let u_* maximize Δ_u, let w carry the largest initial key, and retain the incumbent b and remaining maximum key q_{heap} from Algorithm 1. At a cap stop w has been processed, so the incumbent satisfies $b \geq \Delta_w \geq \text{key}(w)/\chi \geq \Delta_{u_*}/\chi^2$. At an acceptance stop, processed values are at most b and an unprocessed u has $\Delta_u \leq \chi q_{\text{heap}} \leq \chi b/(1-\theta)$; hence $b \geq (1-\theta)\Delta_{u_*}/\chi$.

Greedy recurrence. By monotonicity and supermodularity of the hard-CFCC cost [1], the gap to OPT_k is at most k times the largest marginal, so it contracts by $1-\alpha_{\text{step}}/k$ per round, giving $1-(1-\alpha_{\text{step}}/k)^k \geq 1-e^{-\alpha_{\text{step}}}$. Finally $1-\chi^{-2} \leq 4\zeta$, $1-(1-\theta)\chi^{-1} \leq \theta+2\zeta$, and $e^{-\alpha_{\text{step}}}-e^{-1} \leq 1-\alpha_{\text{step}}$, while $\mathcal{N}_{N,k} \leq (k+1)N^{k+1}$ yields the dimension order. □

The sufficient dimension records the price of adaptive reuse. Fresh independent solution sketches require $d_{\text{fresh}} = \lceil 8\zeta^{-2}\log(4Nk/\delta)\rceil$ for the corresponding ideal-solve event: each current matrix is fixed conditional on preceding rounds. Maintained solution sketches instead cover all feasible matrices at initialization. This extra dependence on k trades a larger uniform-accuracy requirement for fewer repeated solves.

The bound is a worst-case sufficient condition. For example, $\varepsilon = 0.5$, $\zeta = \varepsilon/8$, $\delta = 0.1$, and $k = 19$ give $d \geq 445{,}823$ at $N = 317{,}079$ and $d \geq 492{,}870$ at $N = 10^6$. Our experimental $d = 64$ is an empirical operating point, outside these sufficient-dimension settings; Table VIII evaluates its accuracy–work tradeoff.

B. Numerical accuracy

Practical solves perturb both update paths. For a computed solution sketch $\widetilde{Y}$ and its ideal counterpart Y, write $\mathscr{E} = \widetilde{Y} - Y$ and measure the error by $\|\mathscr{E}\|_{F,A_0} = \sqrt{\text{tr}(\mathscr{E}A_0\mathscr{E}^\top)}$. Let e_1, e_2 bound this norm for the first and second solution sketches. Let $a > 0$ and $B > 0$ bound the smallest and largest eigenvalues of A_0, respectively. The next result propagates a selected-column residual into solution sketch and moment errors. Supported errors and columns have zero coordinates at already grounded vertices.

Proposition 3 (Residual-controlled maintenance)**.** *Let $a \leq \lambda_{\min}(A_0)$ and $B \geq \lambda_{\max}(A_0)$. Suppose a supported selected column $\widetilde{x}$ has $\widetilde{x}_v > 0$ and true active-system residual norm at most ρ_{res}. If a solution sketch's current error is at most e in $\|\mathscr{E}\|^2_{F,A_0} = \text{tr}(\mathscr{E}A_0\mathscr{E}^\top)$, its update has error at most*

$$e^+ = e + \frac{\|\widetilde{Y}[:,v]\|\rho_{\text{res}}}{\sqrt{a}\,\widetilde{x}_v} + \omega, \tag{9}$$

where ω bounds supported rounding error in the same norm. On the ideal-sketch event with tolerance ζ_0, the relative moment errors are bounded by

$$\zeta_0 + 2\sqrt{1+\zeta_0}\,e_1 + e_1^2, \quad \zeta_0 + 2\sqrt{1+\zeta_0}\sqrt{B}\,e_2 + Be_2^2.$$

Thus a target tolerance $\bar{\zeta} < 1$ is met by $\zeta_0 = \bar{\zeta}/2$ and $e_1, \sqrt{B}e_2 \leq \bar{\zeta}/8$.

Proof sketch. Let U be the active coordinates, $x = R_T e_v$, $h = \widetilde{x} - x$, and $\|w\|^2_{A_0} = w^\top A_0 w$. The residual gives $\|h\|_{A_0} \leq \rho_{\text{res}}/\sqrt{a}$. Active inverse equations give $\langle h, x\rangle_{A_0} = h_v$ and $\|x\|^2_{A_0} = x_v$; hence

$$\left\| \frac{\widetilde{x}}{\widetilde{x}_v} - \frac{x}{x_v} \right\|^2_{A_0} = \frac{\|h\|^2_{A_0} - h_v^2/x_v}{\widetilde{x}_v^2} \leq \frac{\rho^2_{\text{res}}}{a\widetilde{x}_v^2}.$$

Put $P = I - e_v x^\top/x_v$ and $\mathscr{E} = \widetilde{Y} - Y$. Expanding the supported-row energies yields

$$\|\mathscr{E}P\|^2_{F,A_0} = \|\mathscr{E}\|^2_{F,A_0} - \|\mathscr{E}[:,v]\|^2/x_v.$$

The computed-minus-ideal update is

$$\begin{aligned}\widetilde{Y}' - Y' = \mathscr{E}P + D_{\text{rnd}} \\ + \widetilde{Y}[:,v](x/x_v - \widetilde{x}/\widetilde{x}_v)^\top,\end{aligned}$$

where $\|D_{\text{rnd}}\|_{F,A_0} \leq \omega$. Contraction of $\mathscr{E}P$, the outer-product norm, and the triangle inequality give (9). For sketch j, the coordinate-representer bound is $\|\mathscr{E}[:,u]\| \leq e_j\sqrt{r_1(u)}$. For Y_1 this is relative to $\sqrt{r_1}$; for Y_2 use $r_1 \leq Br_2$. Expanding the perturbed squared norms on the ideal-sketch event gives the two stated moment-error bounds. □

Retained scalar moments follow a separate error path: their refresh error plus accumulated decrement-error bounds. The coefficients use independently solved selected columns, so candidate-cache errors do not feed back into them. To combine both paths, let $\nu < 1$ bound the relative error of computed refreshed marginals and let $\bar{\zeta} < 1$ be the target solution sketch-moment tolerance. Define

$$\eta = \max\left\{ \frac{1+\bar{\zeta}}{1-\bar{\zeta}} - 1, \nu \right\} < 1.$$

A sufficient inexact step factor is

$$\alpha_{\text{num}} = \frac{1-\nu}{1+\nu}\min\left\{ \frac{1-\eta}{1+\eta}, (1-\theta)(1-\eta) \right\}.$$

The minimum follows the two stopping cases in Theorem 2; the prefactor accounts for choosing the incumbent using approximate refreshed values. Assigning sampling and numerical budgets so that $\alpha_{\text{num}} \geq 1-\varepsilon$ gives the stated $(1-1/e-\varepsilon)$ seeded-gain factor. Thus sketch concentration, column residuals, and retained-moment errors enter one approximation condition.

C. Solver implementation

GROVE constructs an approximate-Cholesky seed preconditioner and masks grounded coordinates in later applications. Blocks with fewer than four right-hand sides use Jacobi preconditioning; on roadNet-PA the seed factor is used for every block. This mixed solver policy is part of the evaluated implementation.

For work accounting, suppose the seed approximate inverse B_0 satisfies $a_c A_0^{-1} \preceq B_0 \preceq b_c A_0^{-1}$, with $0 < a_c \leq b_c$, and put $\chi_c = b_c/a_c$. Here $P \preceq Q$ means $Q - P$ is positive semidefinite. For active coordinates U after t additions, define $A = A_0[U,U]$ and $B_U = B_0[U,U]$. The preconditioned conjugate-gradient

(PCG) error is measured in $\|e_h\|_A = \sqrt{e_h^\top A e_h}$, where e_h is the solution error after h iterations. Under the stated seed-quality premise, $t + O(\sqrt{\chi_c}\log(1/\xi))$ iterations suffice for relative energy accuracy ξ, capped by $|U|$. The block inverse formula gives $(A_0^{-1})[U,U] = A^{-1} + ZZ^\top$, where the positive Schur-complement correction has $\operatorname{rank} Z \le t$. After congruence by $A^{1/2}$, restricting the seed bounds shows that $A^{1/2}B_U A^{1/2}$ has no eigenvalue below a_c and at most t above b_c. A PCG residual polynomial places roots at the large outliers and uses a Chebyshev factor on $[a_c, b_c]$. The outlier factors have magnitude at most one on that interval. For $\chi_c > 1$ and $h \ge t$, this gives

$$\frac{\|e_h\|_A}{\|e_0\|_A} \le 2\left(\frac{\sqrt{\chi_c}-1}{\sqrt{\chi_c}+1}\right)^{h-t},$$

which yields the iteration bound. For $\chi_c = 1$, at most $t+1$ iterations suffice; the dimension cap is $|U|$ in either case. The Kyng–Sachdeva construction [3] supplies one admissible seed-quality guarantee. The implementation uses a lightweight clique-sampling variant, with separate condition-number-dependent bounds for its Jacobi calls.

D. Work and comparison with fresh-round selection

The work saved by reuse depends on how often solution sketch updates and retained values replace candidate solves. Let T_{build} be preconditioner construction work, n_{ref} the total number of distinct refreshed candidates, n_t^{ref} the count in round t, F_c the number of stored factor entries, and $\mathscr{T}_t$ the work of one solve at the required precision in that round. The hybrid's work is

$$T_{\text{build}} + O\Bigg(dm + 2d\mathscr{T}_0 + \sum_{t=0}^{k-1} n_t^{\text{ref}}\mathscr{T}_t + 2\sum_{t=0}^{k-2}\mathscr{T}_t + kNd + kN + (kn_{\text{ref}} + n_{\text{ref}} + k)\log(N+1)\Bigg). \tag{10}$$

The dm term forms the incidence-based sketch right-hand sides, and $2d\mathscr{T}_0$ covers initial solution sketch solves; the two sums cover candidate refreshes and selected-column maintenance. Reusing the winning column reduces the latter. The kNd and kN terms cover solution sketch and scalar updates, respectively. The solution sketches require $O(Nd)$ storage in addition to the graph, factor, scalar moments, and solve batches. Exact retention makes $n_{\text{ref}} \le N$; with a common positive cap and batch width, $n_{\text{ref}} \le \min\{N, k(b_r + B_r - 1)\}$. The heap term covers previously refreshed candidates that are processed again without a solve.

Under the preceding seed-quality premise, an all-factor configuration can take

$$\mathscr{T}_t = O\big((m+N+F_c)\min\{N-t, t+1+\sqrt{\chi_c}\log(2/\xi)\}\big).$$

Jacobi calls instead use their own condition-number-dependent iteration bound. The maintained algorithm is nearly linear in graph size for fixed budget, precision, and refresh parameters. Substituting the uniform dimension into (10) exposes an array-update term of order $\widetilde{O}(k^2 N\varepsilon^{-2})$, alongside initialization and selected/refreshed solves.

At $s = 1$, fresh-round JL selection with an SDDM solver has $\widetilde{O}(km\varepsilon^{-2})$ selection work, as in ApproxGreedy [1], [12]. This uses fresh $O(\varepsilon^{-2}\log(Nk/\delta))$-row sketches and accurate solves in each round. In contrast, (10) retains the maintained method's explicit budget, array-update, and refresh costs; empirical savings arise when updates replace enough solves.

V. Extension: Completely Monotone Objectives

The CFCC algorithm uses only $s = 1$ and the known hard-grounding structure [1]. This section extends that structure to every CM trace objective and supports the separate inverse-power experiment in Section VI-G. Positive integer powers $p \ge 1$ provide a tunable objective for exact-marginal selection on medium graphs, allowing aggregate resistance and slow modes to receive different weights.

A. CM grounding structure

The *admissible lattice* is the family of sets containing the fixed seed S_0; an *available vertex* lies outside the current selected set.

Theorem 4 (CM grounding structure)**.** *For every CM function f and every $\beta \in (0,\infty]$, Φ_f is nonincreasing and supermodular on the admissible lattice. At an available vertex, a soft-grounding drop is strictly positive exactly when f is nonconstant; a hard-grounding drop is strictly positive exactly when $f \not\equiv 0$. Writing $a_f = \lim_{x\to\infty} f(x) = \mu(\{0\})$, the limit is*

$$\lim_{\beta\to\infty} \operatorname{tr} f(A_S^\beta) = \operatorname{tr} f(A_S^\infty) + |S| a_f.$$

In particular, inverse powers and heat with $\tau > 0$ converge directly to the hard-grounding objective.

Proof sketch. Couple all sets through one continuous-time walk X_t with jump rates w_{ij} (Fig. 1(a)). For finite β, draw independent exponential clocks E_0, E_v of rates σ, β, taking $E_0 = \infty$ if $\sigma = 0$. Define occupation time $\ell_v(t) = \int_0^t \mathbf{1}\{\mathsf{X}_u = v\}\,du$, $\kappa_v = \inf\{t : \ell_v(t) \ge E_v\}$, and killing time $\mathsf{T}_S = \min\{E_0, \min_{v\in S}\kappa_v\}$. For hard grounding, κ_v is the first hitting time of v. Then $\mathsf{T}_{S\cup T} = \min\{\mathsf{T}_S, \mathsf{T}_T\}$ pathwise. Conditional soft survival is $e^{-\sigma t - \beta\sum_{v\in S}\ell_v(t)}$, giving generator $-A_S^\beta$. Hard absorption gives the principal-matrix semigroup, embedded by zeros on S.

For a walk starting at i, put $Z_S = \mathbf{1}\{\mathsf{X}_t = i,\ t < \mathsf{T}_S\}$. If $S \subseteq T$ and $v \notin T$, then $Z_S \ge Z_T$ and

$$\begin{aligned} Z_S - Z_{S\cup\{v\}} &= Z_S\mathbf{1}\{\kappa_v \le t\} \\ &\ge Z_T\mathbf{1}\{\kappa_v \le t\} = Z_T - Z_{T\cup\{v\}}. \end{aligned}$$

Expectations summed over i give the claims for $h_S^\beta(t) = \operatorname{tr} e^{-tA_S^\beta}$. The finite traces and Tonelli's theorem let these nonnegative drops be integrated against μ in (2).

For strictness, the no-jump event from v contributes $e^{-(d_v+\sigma)t}(1-e^{-\beta t})$ to a soft drop and $e^{-(d_v+\sigma)t}$ to a hard drop, where $d_v = \sum_j w_{vj}$. The hard drop at $t = 0$ is one. Thus the soft case is strict exactly when $\mu((0,\infty)) > 0$, or f is nonconstant; the hard case is strict exactly when $\mu \ne 0$.

For the limit, use common unit-rate clocks and set $E_v^\beta = E_v^1/\beta$. At every $t > 0$, survival converges almost surely to

hard survival, so bounded convergence gives $h_S^{\beta}(t) \to h_S^{\infty}(t)$. For fixed $\beta_0 > 0$, $h_S^{\beta_0}$ is an integrable upper bound for $\beta \geq \beta_0$. At $t = 0$ the soft and hard traces are n and $n - |S|$, respectively. Dominated convergence therefore gives the stated limit, including the atom $|S|\mu(\{0\})$. □

B. Integer marginal evaluation

For an active inverse R and candidate v, put $x = Re_v$, $c = 1/r_1(v)$, and $\bar{R}_v = R - cxx^\top$. The marginal $\Delta_p = \operatorname{tr} R^p - \operatorname{tr} \bar{R}_v^p$ can be evaluated from moments $r_j(v) = e_v^\top R^j e_v$, avoiding separate traces. Set $a_j = cr_{j+1}(v)$ and let g_j be the degree-j coefficient of $\log(r_1 - r_2 z_{\mathrm{ser}} + r_3 z_{\mathrm{ser}}^2 - \cdots)$, with formal series variable z_{ser}. Then $\Delta_p = (-1)^p p g_p$, so moments through $p+1$ suffice. For example,

$$\Delta_2 = 2r_3/r_1 - (r_2/r_1)^2. \tag{11}$$

The following identity exposes the nonnegative terms behind such alternating expressions.

Proposition 5 (Integer marginal identity)**.** *For $j \geq 1$, define $d_j = cx^\top \bar{R}_v^{j-1} x \geq 0$, $H_0 = 1$, and $H_j = \sum_{i=1}^{j} d_i H_{j-i}$. For every integer $p \geq 1$, the inverse-power marginal is*

$$\Delta_p = \operatorname{tr} R^p - \operatorname{tr} \bar{R}_v^p = \sum_{j=1}^{p} j d_j H_{p-j}.$$

All terms are nonnegative, and moments through $r_{p+1}(v)$ determine the marginal.

Proof. For the formal variable z_{ser}, define $\mathscr{D}(z_{\mathrm{ser}}) = \sum_{j \geq 1} d_j z_{\mathrm{ser}}^j$. Since $R = \bar{R}_v + cxx^\top$, the determinant lemma and the resolvent series give, near zero,

$$\frac{\det(I - z_{\mathrm{ser}} \bar{R}_v)}{\det(I - z_{\mathrm{ser}} R)} = \frac{1}{1 - \mathscr{D}(z_{\mathrm{ser}})},$$

$$\sum_{p \geq 1} \frac{\Delta_p}{p} z_{\mathrm{ser}}^p = -\log(1 - \mathscr{D}(z_{\mathrm{ser}})).$$

The coefficients of $H(z_{\mathrm{ser}}) = (1 - \mathscr{D}(z_{\mathrm{ser}}))^{-1}$ obey the stated recurrence. Differentiating the logarithmic identity and comparing coefficients gives $\Delta_p = \sum_{j=1}^{p} j d_j H_{p-j}$. Positive semidefiniteness of $\bar{R}_v$ gives $d_j, H_j \geq 0$. Applying the determinant lemma in the reverse direction also gives

$$H(z_{\mathrm{ser}}) = 1 + \sum_{j \geq 1} c r_{j+1}(v) z_{\mathrm{ser}}^j.$$

Thus coefficients through degree p require only the stated moments. □

The power experiments obtain the required moments by successive linear solves and use numerical exact-marginal selection.

C. From trace weighting to the smallest eigenvalue

For a symmetric positive-definite (SPD) grounded matrix, the inverse trace combines all eigenvalues, whereas the smallest eigenvalue describes the slowest mode. Their connection depends on the guarantee available for selection. A trace-cost factor $\alpha_{\mathrm{cost}} \geq 1$ compares final costs; a seeded-gain factor $\gamma \in [0,1]$ compares reductions from the initial cost $\Phi_s^0 = \Phi_s(S_0)$. Let $D = n - |S_0| - k$ be the final active dimension, λ_k^* the largest attainable smallest eigenvalue among extensions of this seed, and X_{out} the output's grounded matrix. The next result keeps the baseline dependence explicit.

Proposition 6 (A baseline-aware spectral bridge)**.** *For an SPD matrix X of dimension D and every $s > 0$,*

$$\lambda_{\min}(X)^{-1} \leq (\operatorname{tr} X^{-s})^{1/s} \leq D^{1/s} \lambda_{\min}(X)^{-1}.$$

At a fixed final dimension D, an α_{cost}-approximation to minimum trace, $\alpha_{\mathrm{cost}} \geq 1$, has spectral factor $(\alpha_{\mathrm{cost}} D)^{-1/s}$ relative to the largest attainable smallest eigenvalue λ_k^*. If instead the available guarantee is gain factor γ, write $\Phi_s^0 = \Phi_s(S_0)$ and $\rho_{\mathrm{gain}} = 1 - \gamma$. Then

$$\frac{\lambda_{\min}(X_{\mathrm{out}})}{\lambda_k^*} \geq [\rho_{\mathrm{gain}} \Phi_s^0 (\lambda_k^*)^s + \gamma D]^{-1/s}.$$

The inverse-power root approaches the reciprocal smallest eigenvalue uniformly, with relative excess at most $e^{\log D/s} - 1$.

Proof sketch. Factor $\lambda_{\min}^{-s}$ from the trace; the remaining D terms are at most one and at least one equals one. For the gain bound, use $\Phi_s(\mathrm{out}) \leq \rho_{\mathrm{gain}} \Phi_s^0 + \gamma \min \Phi_s$ and $\min \Phi_s \leq D(\lambda_k^*)^{-s}$. Combining the two inequalities yields the baseline-dependent spectral factor. □

The bound explains how inverse powers approach the spectral objective, while retaining the distinction between final-cost and gain guarantees. A finite grounding decision can therefore rank vertices differently from a leading-eigenvector sensitivity score. Zhou et al. [6] use the Fast score $h_v = 2\phi_1(v) \sum_{u \sim v, u \notin S} \phi_1(u)$ for unit weights, where ϕ_1 is a unit eigenvector for the smallest grounded eigenvalue and $u \sim v$ denotes adjacency. In comparison, define the finite-removal path $Q_p(\vartheta) = \operatorname{tr} R^p - \operatorname{tr}(R - \vartheta cxx^\top)^p$ for local interpolation parameter $\vartheta \in [0,1]$. The endpoints represent no removal and complete removal, respectively. Its initial sensitivity is

$$Q_p'(0) = \frac{p}{r_1(v)} \sum_i \lambda_i^{-(p+1)} \phi_i(v)^2,$$

where λ_i, ϕ_i are the current grounded eigenvalues and orthonormal eigenvectors. Thus all modes contribute, with smaller eigenvalues weighted more strongly. At $p = 2$, $Q_2(\vartheta) = 2\vartheta a_2 - \vartheta^2 a_1^2$ also includes a finite-removal correction at $\vartheta = 1$, as follows by expanding the squared updated inverse and taking its trace. These differences motivate evaluating both inverse trace and smallest eigenvalue as p varies, separately from testing the efficiency of the large-scale $p = 1$ algorithm.

VI. Experimental Evaluation

We test whether reusing current sketches and retained moments reduces selection cost while preserving CFCC quality. The comparisons measure this tradeoff at scale; runtime components locate saved and remaining work; ablations isolate the two maintenance paths; parameter and seed studies examine when the savings persist. Finally, inverse-power experiments test the spectral weighting that motivated the broader objective family.

TABLE II

MAIN COMPARISON: HARD CFCC, $K = 20$, SELECTION $d = 64$, 14 THREADS, 128 COMMON EVALUATION PROBES. EACH METHOD HAS SELECTION TIME IN SECONDS AND ESTIMATED $\widehat{\Phi}_1 \approx \mathrm{tr}X^{-1}$ (LOWER IS BETTER). ALL METHODS START FROM THE SAME COMMON SEED; SCHURCFCM AND FORESTCFCM DRAW AT MOST 1400 FORESTS PER ROUND. FORESTCFCM WAS NOT RUN ON LIVEJOURNAL.

	GROVE		ApproxGreedy		SchurCFCM		ForestCFCM		Zhou	
Graph	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$
DBLP	42	188268.0	115	188365	419	188390.6	392	188821.6	16	188716
Amazon	143	221167.8	183	221435	468	221003.3	475	222317.3	144	221917
YouTube	113	1091405.9	646	1091648	1666	1091368.5	1639	1094043.2	852	1091395.5
web-Google	469	479082.5	506	479486.8	995	479384.9	1025	482329.8	1013	479743.9
Skitter	297	676053.0	1378	676081.8	2879	676487.6	2239	678054.8	6124	676009.8
roadNet-PA	356	3.2230M	795	3.275M	2461	3.1992M	2699	3.2060M	3526	3.245M
LiveJournal	4626	1626554	> 5400	–	> 7200	–	–	–	> 3600	–

TABLE III

MAIN DATASETS: LARGEST CONNECTED COMPONENTS AFTER PREPROCESSING.

Graph	n	m
com-DBLP	317,080	1,049,866
com-Amazon	334,863	925,872
com-YouTube	1,134,890	2,987,624
web-Google	855,802	4,291,352
as-Skitter	1,694,616	11,094,209
roadNet-PA	1,087,562	1,541,514
com-LiveJournal	3,997,962	34,681,189

A. Datasets and protocol

a) Graphs and budget: The seven SNAP graphs [15] have the component sizes listed in Table III. Inputs are unweighted and symmetrized, with loops and duplicate edges removed, then stored in Cuthill–McKee order. Every vertex outside the common singleton seed is eligible. We use hard grounding, $\sigma = 0$, and $K = 20$ total anchors unless stated otherwise. The seed-stage Johnson–Lindenstrauss (JL) dimension is 128; selection uses $d = 64$, the empirical setting studied in Table VIII rather than the worst-case sufficient size of Theorem 2.

b) Compared methods: All baselines are our C++ reimplementations inside one engine. They share GROVE's graph storage, PCG solver, preconditioner, and thread settings, so same-machine time differences reflect the algorithms rather than languages or solver libraries; we located no released code for the forest methods [5]. ApproxGreedy is fresh-round all-node JL marginal estimation [1]. Its released code[1] accompanies the original publication and targets Julia 0.6 with the Laplacians.jl solver; we therefore reimplement it in the shared engine with JL dimension 64, the same as GROVE's d, and compare its marginals and selected sets with dense exact greedy on small graphs. ForestCFCM and SchurCFCM are Xia and Zhang's forest-sampling algorithms [5]. SchurCFCM was checked against dense exact values: fixed-set estimates converge over up to 10^5 forests, nine $k = 10$ selections have exact traces within 0.81% of exact greedy, and its auxiliary-root counts on DBLP, Amazon, and Skitter (131, 96, 965) equal those reported in [5]. Both forest methods use JL dimension 64 and input parameter $\varepsilon_{\mathrm{forest}} = 0.2$. Their adaptive stopping rule did not terminate on these graphs, so each round draws at most 1400 forests, 100 per thread; a fourfold larger cap on DBLP leaves the comparison unchanged (Section VI-B). SchurCFCM uses the original degree-based auxiliary-root rule, producing between 8 auxiliary vertices on roadNet-PA and 965 on Skitter. Zhou is the Fast score [6] with warm-started inverse power iteration targeting $\lambda_{\min}$. All selectors receive the same initial anchor.

c) Execution and parameters: All methods run on one host with two Intel Xeon Gold 6330 CPUs at 2.0 GHz, 755 GB RAM, and Ubuntu 22.04. Each job is limited to 14 cores and 14 OpenMP threads. The C++17 engine is compiled by g++ 11.4.0 with `-O3 -march=native -std=c++17 -fopenmp`. Selection time includes factor/sketch construction, refinement, and updates; it excludes graph loading, the shared seed stage, and independent evaluation. The seed stage takes 3.747–62.708 seconds.

GROVE uses both update paths, selected-column reuse, and acceptance tolerance $\theta = 0$. Initialization and refresh batch widths are $B_{\mathrm{init}} = 32$ and $B_r = 16$. The refresh cap b_r is 16 on roadNet-PA and 64 elsewhere. On Skitter no round exceeds 32 refreshes, so this cap is equivalent to unlimited refreshes. The seed preconditioner uses approximate Cholesky as described in Section IV; solve tolerances are $\varepsilon_{\mathrm{init}} = 10^{-3}$, $\varepsilon_{\mathrm{ref}} = 10^{-6}$, and $\varepsilon_{\mathrm{upd}} = 10^{-9}$.

d) Evaluation: Let X be the final active grounded matrix. We estimate $\Phi_1 = \mathrm{tr}X^{-1}$ and $\mathrm{tr}X^{-2}$ with 128 common Rademacher probes [13] and compute $\lambda_{\min}(X)$ by Lanczos on X^{-1}. Evaluation solves use relative residual tolerance 10^{-7}. Lower estimated inverse trace $\widehat{\Phi}_1$ means higher CFCC. For a method pair, the paired gap is $\widehat{\Phi}_1(\mathrm{other}) - \widehat{\Phi}_1(\mathrm{GROVE})$, positive when GROVE is better. Its standard error (SE) measures evaluation noise, and $|\mathrm{gap}|/\mathrm{SE}$ is the standardized gap. We use $|\mathrm{gap}|/\mathrm{SE} > 2$ as a descriptive threshold for a detectable difference. Incomplete runs have a time limit but no final objective value.

B. Main comparison

Reusing state reduces selection time with small objective differences. On the six graphs with completed main comparisons, every baseline's inverse trace is within 1.6% of GROVE's, while times differ substantially (Table II, Fig. 2).

[1] https://github.com/lchc/CFCC-maximization

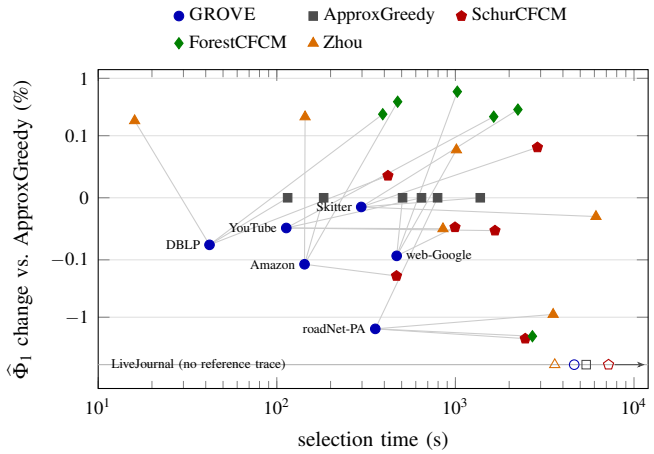


Fig. 2. Selection time against the relative change of estimated inverse trace with respect to ApproxGreedy on the same graph (negative is better); gray segments join GROVE to each baseline on the same graph. The vertical axis is symmetric-logarithmic: linear within $\pm 0.01\%$, logarithmic beyond. On LiveJournal (bottom strip), hollow markers give the GROVE time and the censored baseline limits; the arrow marks censoring.

a) ApproxGreedy: The fresh-round comparison directly tests the cost of rebuilding estimates. GROVE is 1.1–5.7× faster on all six graphs: 5.7× on YouTube, 4.6× on Skitter, 2.7× on DBLP, and 1.1× on web-Google. Over four further selection seeds on four graphs (Table VI), the ratio of mean times is 1.4–3.8×. The YouTube main run (113 seconds) is faster than GROVE's multi-seed mean of 156 seconds, so its multi-seed ratio is 3.75× rather than 5.7×. GROVE's estimated inverse trace is lower on all six, with paired gaps exceeding two SE on five; Skitter's gap is 28.8 ± 22. The largest difference is on roadNet-PA: 3.2230×10^6 in 356 seconds versus 3.275×10^6 in 795 seconds.

b) Forest sampling: GROVE is 2.1–14.8× faster than the two forest methods under the 1400-forest cap. ForestCFCM has higher inverse traces on five graphs, but a 0.53% lower trace on roadNet-PA. SchurCFCM is lower on Amazon and roadNet-PA by 0.07% and 0.74%, at 3.3× and 6.9× the time; its YouTube difference is within two SE. It is higher on DBLP, web-Google, and Skitter. On DBLP, raising the forest budget to 5600 gives SchurCFCM a gap of -24.6 ± 11.5 in 905 seconds: a small 0.01% improvement slightly beyond two SE. ForestCFCM reaches $+235.7 \pm 24.4$ in 1591 seconds. A fourfold forest budget therefore changes neither the time ordering nor the scale of the quality differences.

c) Zhou: The spectral heuristic tests the effect of optimizing a different objective. Zhou is faster on DBLP (16 versus 42 seconds) and almost equally fast on Amazon (144 versus 143). GROVE is faster on the other four graphs, by 2.2–20.6×. Its inverse trace is lower on DBLP, Amazon, web-Google, and roadNet-PA. YouTube is within two SE, while Zhou is slightly better on Skitter (-43.1 ± 19.6, or 0.006% of the trace).

d) LiveJournal: On the largest input, GROVE finishes in 4626 seconds with estimated inverse trace 1626554. ApproxGreedy reaches its 5400-second limit during round 17, SchurCFCM its 7200-second limit after 14 of 19 rounds, and Zhou its 3600-second limit after three rounds. The limits of 90, 120, and 60 minutes were fixed per method before the runs, from pilot per-round costs on the shared server. The ApproxGreedy and SchurCFCM limits exceed GROVE's completion time, so both are slower on this input whatever the limit; Zhou's second round alone took 3140 seconds. The fixed-forest ForestCFCM batch covered the other six graphs, so its LiveJournal cell is unavailable. The scalar-cache ablation completes separately.

TABLE IV
MAIN-RUN RUNTIME COMPONENTS (SECONDS) AND SPMV COUNTS. SELECT IS TOTAL MINUS INIT AND UPDATE, NOT A SEPARATELY TIMED REFRESH KERNEL. G DENOTES GROVE AND AG DENOTES APPROXGREEDY; – IS AN INCOMPLETE BASELINE.

	GROVE time			SpMVs	
Graph	Init	Select	Update	G	AG
DBLP	5.7	25.7	10.7	15185	32672
Amazon	7.5	114.4	21.5	45766	55168
YouTube	20.1	69.0	23.7	13053	49856
web-Google	26.5	383.9	58.9	49618	48192
Skitter	41.6	144.8	110.7	18357	51872
roadNet-PA	36.3	259.1	60.4	23433	69536
LiveJournal	220.9	3856.0	549.1	43599	–

C. Runtime components and solve work

Table IV separates initialization, updates, and the remaining selection work. Initialization includes factor construction and sketch solves; updates include selected-column solves and sketch/scalar maintenance. The selection remainder includes refreshes, heap operations, and other overhead. Sparse matrix–vector products (SpMVs) are counted per right-hand side, not as batched PCG iterations.

The profiles expose distinct regimes. YouTube uses 13,053 SpMVs versus 49,856 for ApproxGreedy. On Skitter, update work already takes 110.7 seconds, or 37.3% of the total. On web-Google, GROVE uses 49,618 SpMVs versus 48,192 for ApproxGreedy: 640 candidate refreshes and three cap hits leave 383.9 seconds in the selection remainder (81.8% of runtime), consistent with the small 1.08× speedup. Solves of different widths and tolerances have different costs, so these counts describe workload rather than predict elapsed time.

On roadNet-PA every round reaches the cap at $K = 10, 20, 50$, giving 144, 304, and 784 refreshes. Per-round refresh work therefore persists as the budget grows. The speedup rises modestly from 2.08× to 2.54×; it is the paired quality gap, not the speed ratio, that narrows at $K = 50$ (Table VII).

D. Two complementary maintenance paths

Table V compares the two reuse paths on a common set of rows. Scalar caching starts from noisy sketch moments and retains their initial absolute errors under exact decrements. GROVE instead keeps current solution sketches for unrefreshed candidates and retains solved moments for refreshed candidates. The ablation measures selection time and final quality; Fig. 3 measures the per-candidate errors directly.

Scalar caching is competitive on most graphs. On the six inputs other than roadNet-PA its time is 0.95–1.15× that

TABLE V

UPDATE-PATH ABLATIONS: TIME (S) / ESTIMATED INVERSE TRACE. NA MEANS AN INDEPENDENT SKETCH-ONLY RUN IS UNAVAILABLE; – MEANS A COMPLETED SET WAS NOT EVALUATED. SKETCH-ONLY USES CAP 16 ON ROADNET-PA; ITS YOUTUBE AND SKITTER CAPS ARE NONBINDING. GROVE USES THE MAIN COMPARISON SETTINGS.

Graph	Scalar cache	Sketch-only	GROVE
DBLP	40 / 188268	NA	42 / 188268.0
Amazon	136 / 221259	NA	143 / 221167.8
YouTube	130 / 1091406	424 / 1091405.9	113 / 1091405.9
web-Google	490 / 479041.6	NA	469 / 479082.5
Skitter	288 / 676053.6	570 / 676052.1	297 / 676053.0
roadNet-PA	881 / 3.659M	278 / 3.2366M	356 / 3.2230M
LiveJournal	5206 / –	NA	4626 / 1626554

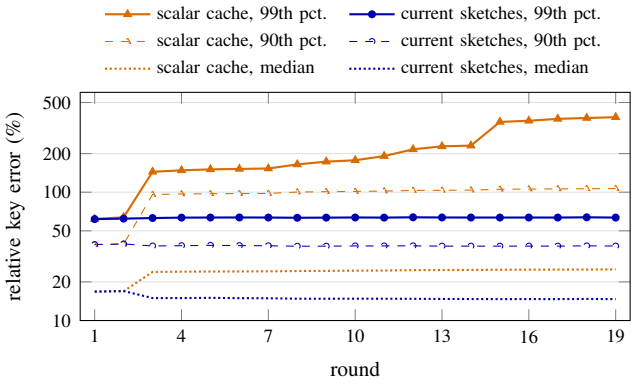


Fig. 3. HepPh, $d = 64$: relative error of the estimated key r_2/r_1 against exact moments before each round, over all active candidates (mean of five projection draws). The scalar cache keeps its initial absolute error; the current sketches used by GROVE for unrefreshed candidates keep a constant relative error.

of GROVE, and its trace is equal within noise on DBLP, YouTube, and Skitter, 0.04% higher on Amazon, and lower on web-Google (paired gap -40.9 ± 3.3). It has no error control, however, and it fails on roadNet-PA: 881 seconds and 3.659×10^6, a trace 13.5% above GROVE's 3.2230×10^6 at 2.5× the time. The value of GROVE is that it avoids this failure without giving up speed on the other graphs.

Fig. 3 shows the mechanism on HepPh ($n = 11{,}204$), where dense inverses give exact moments. Both estimators start from the same $d = 64$ sketches and follow the exact-greedy trajectory from the common seed. The current sketches keep the relative error of the key r_2/r_1 flat: its 90th and 99th percentiles stay at 38% and 63% through all 19 rounds. The scalar cache preserves its initial absolute error exactly while moments near the new anchors shrink, so the same percentiles rise to 107% and 384%. Refreshed GROVE candidates carry exact moments.

Sketch-only removes retention. It takes 424 seconds on YouTube and 570 on Skitter, versus 113 and 297 for GROVE, with nearly identical trace estimates. On Skitter, 12 of 19 hybrid rounds need no refresh; only 128 candidates are ever refreshed. Independent sketch-only results cover these two graphs and roadNet-PA; missing cells are shown explicitly. LiveJournal's completed scalar-cache set was not evaluated. The cap-64 roadNet-PA sketch-only result is discussed below.

TABLE VI

FOUR FURTHER SEEDS OF THE SELECTION RANDOMNESS, $K = 20$, $d = 64$, FIXED COMMON SEED VERTEX: MEAN ± STANDARD DEVIATION OF SELECTION TIME (S) AND OF THE PAIRED GAP $(\widehat{\Phi}_1(\text{APPROXGREEDY}) - \widehat{\Phi}_1(\text{GROVE}))/\widehat{\Phi}_1$; THE LAST COLUMN GIVES THE RANGE OF PER-SEED PAIRED GAP/SE RATIOS.

Graph	GROVE	ApproxGreedy	Speedup	Gap (%)	Gap/SE
DBLP	37.6 ± 3.4	123.9 ± 2.3	3.29×	0.035 ± 0.009	3.4–5.5
Amazon	136.2 ± 9.2	189.6 ± 8.8	1.39×	0.219 ± 0.047	5.4–10.0
YouTube	156.0 ± 19.6	585.5 ± 41.8	3.75×	0.019 ± 0.010	0.7–8.1
roadNet-PA	343.5 ± 7.9	793.6 ± 18.8	2.31×	1.191 ± 0.078	5.8–7.5

TABLE VII

TOTAL BUDGET K ($d = 64$): SELECTION TIME (S) AND PAIRED DIFFERENCE $\widehat{\Phi}_1(\text{APPROXGREEDY}) - \widehat{\Phi}_1(\text{GROVE}) \pm \text{SE}$; POSITIVE VALUES FAVOR GROVE.

Graph	K	GROVE	ApproxGreedy	Speedup	Difference
DBLP	10	24.3	49.2	2.0×	$+23.8 \pm 12.2$
	20	42.1	115	2.7×	$+97.2 \pm 16.7$
	50	92.7	402.6	4.3×	$+160.0 \pm 17.3$
YouTube	10	56.1	237.4	4.2×	$+65.5 \pm 32.7$
	20	112.9	646	5.7×	$+241.7 \pm 44.3$
	50	346.7	2260.3	6.5×	$+349.8 \pm 43.5$
roadNet-PA	10	160.1	333.6	2.1×	$+48575 \pm 10477$
	20	355.8	795	2.2×	$+51513 \pm 5865$
	50	1020.4	2590.4	2.5×	$+18562 \pm 3366$

E. Sensitivity to budget, dimension, and refresh cap

We vary K, d, and b_r to test the balance between reuse, estimate quality, and refinement work. Tables VII and VIII vary one parameter at a time on DBLP, YouTube, and roadNet-PA; refresh totals count distinct candidates.

a) Budget: More rounds increase the opportunity for reuse. From $K = 10$ to $K = 50$, the speedup over ApproxGreedy rises from 2.0× to 4.3× on DBLP and 4.2× to 6.5× on YouTube; it remains 2.1–2.5× on roadNet-PA. GROVE has lower estimated inverse trace at every tested budget. The gaps exceed four SE except at $K = 10$ on DBLP and YouTube, where they are about two SE.

b) Sketch dimension: A narrower solution sketch can increase refinement cost. Raising d from 16 to 64 reduces YouTube's refreshes from 784 to 144 and time from 407 to 113 seconds; DBLP shows the same pattern. At $d = 128$, trace differences from $d = 64$ are within two SE on all three graphs. On roadNet-PA the cap binds throughout the sweep, and time

TABLE VIII
SELECTION SKETCH DIMENSION d ($K = 20$): SELECTION TIME (S), DISTINCT REFRESHED CANDIDATES, AND PAIRED DIFFERENCE $\widehat{\Phi}_1(d) - \widehat{\Phi}_1(64) \pm$ SE.

	$d = 16$	$d = 32$	$d = 64$	$d = 128$
DBLP time	56.9	46.3	42.1	38.7
refreshes	512	368	256	176
difference	$+50.7 \pm 8.8$	$+4.3 \pm 10.4$	–	-4.7 ± 9.2
YouTube time	406.7	240.8	112.9	115.6
refreshes	784	448	144	96
difference	$+254.9 \pm 45.9$	$+160.1 \pm 33.3$	–	$+21.3 \pm 39.5$
roadNet-PA time	314.8	324.2	355.8	365.1
refreshes	304	304	304	304
difference	$+58297 \pm 6725$	$+22505 \pm 5047$	–	-9086 ± 4940

instead rises mildly with dimension, from 315 to 365 seconds.

c) Refresh cap: The cap trades refinement against quality. On roadNet-PA, sketch-only caps 16 and 64 take 278 and 1069 seconds, reaching 3.2366×10^6 and 3.2098×10^6. The hybrid with cap 16 lies between them in quality, at 3.2230×10^6 in 356 seconds. On Skitter the cap is inactive. These cases separate bounded refinement from the benefits of the two update paths.

F. Variability and paired differences

Selection randomness and evaluation noise measure different sources of variation. Table VI uses four further selection seeds (1001–1004) with a fixed initial anchor. Population standard deviations summarize these runs; speedups are ratios of mean times. GROVE remains faster on all four graphs, by 1.39–3.75×, and has lower estimated inverse trace in all 16 graph–seed pairs. Fifteen paired gaps exceed three SE; the remaining YouTube pair has gap/SE $= 0.7$.

Table X evaluates the main-run sets with common probes to reduce noise in the differences. For example, the DBLP comparison with ApproxGreedy has paired SE 16.7, versus 82 for a single trace estimate. These paired errors support the main quality comparisons; the repeated-selection standard deviations describe run-to-run variation separately.

G. The inverse power as a tunable knob

This experiment tests spectral weighting independently of the large-scale $p = 1$ sketch implementation. On five medium graphs, exact-marginal lazy selection uses $p = 1, 2, 3, 4, 8$, with moments computed by numerical solves. Final traces use unit-vector solves at relative residual tolerance 10^{-9}, and smallest eigenvalues use Lanczos (Table IX, Fig. 4).

From $p = 1$ to $p = 4$, $\lambda_{\min}$ increases by 1.51–2.33×, while inverse trace rises only 0.05–0.60%. Among the tested powers, $p = 1$ gives the lowest trace on each graph. The spectral response is not strictly monotone: CAIDA has a 0.4% dip from $p = 2$ to $p = 3$. At $p = 4$, the smallest eigenvalue is within 0.1% of Zhou's on AstroPh and Enron, and 4.5–9.5% lower on the other graphs. At $p = 8$, it exceeds Zhou's on four graphs and is 0.05% lower on Enron. These sets expose the cost of emphasizing slower modes rather than treating either objective as interchangeable with CFCC.

VII. RELATED WORK

a) Group centrality and grounded spectra: Li et al. [1] formulate CFCC, establish optimization hardness and reciprocal-objective supermodularity, and develop exact and randomized greedy algorithms, including ApproxGreedy. Current-flow closeness of single vertices goes back to Brandes and Fleischer [17], and group centralities to Everett and Borgatti [18]. Greedy and local-search maximization has been engineered for shortest-path group closeness and group harmonic centrality [19], [20], group betweenness by sampling [21], and group forest closeness [22]. Group Markov centrality considers related hitting-time group objectives [7]. Zhou et al. [6] instead maximize the smallest grounded eigenvalue, establishing monotonicity and failure of submodularity, and propose the Fast sensitivity score. Aggregate resistance and the slowest spectral mode capture different aspects of a selected group. Our CM formulation places CFCC in a trace family with tunable inverse-power weighting and an explicit connection to the spectral endpoint.

b) Leader selection: In networked control, choosing k noise-free leaders to minimize the steady-state variance of consensus followers is the same grounded inverse-trace problem [23]. Clark et al. [24] prove its supermodularity and analyze greedy selection; Lin et al. [25] give convex relaxations and bounds. These methods use dense inverses and target networks of hundreds to thousands of nodes. The smallest eigenvalue of the grounded Laplacian, which governs convergence to the leaders, is analyzed by Pirani and Sundaram [26].

c) Forest sampling for CFCC: Xia and Zhang [5] estimate CFCC marginals using spanning forests rooted at the selected group and generated by Wilson's algorithm [4]. ForestCFCM estimates marginals directly; SchurCFCM introduces auxiliary high-degree roots and applies Schur-complement corrections. Both draw new forests after each selection and use graph-dependent bounds for adaptive sampling. GROVE instead maintains solution sketches across grounding updates and retains refreshed moments, reducing repeated all-candidate estimation.

d) Spectral design and matrix functions: FastFrac [8] optimizes positive fractional traces under edge addition; POLAR [9] studies negative fractional objectives through submodularity-ratio analysis. Edge modification changes walk transition rates, whereas grounding changes survival by killing or removing active coordinates. Matrix-function representations and quadrature [14] evaluate spectral objectives through resolvents. Our integer formulation uses inverse moments for finite grounding marginals and their incremental updates.

e) Randomized and incremental numerical methods: JL projections reduce inverse-diagonal estimation to a small number of solves, as in effective-resistance sketching [2] and ApproxGreedy [1]. Sherman–Morrison identities [16] give rank-one inverse updates, while classical lazy greedy [11] uses stale submodular upper bounds. GROVE combines current sketches, retained moments, and capped refresh; its analysis handles

TABLE IX

EXACT-MARGINAL LAZY SELECTION ON MEDIUM GRAPHS, $K = 20$, COMMON SEED, $\sigma = 0$. EACH ENTRY IS $\lambda_{\min} / \operatorname{tr} X^{-1}$, EVALUATED EXACTLY: THE TRACE SUMS UNIT-VECTOR SOLVES AT RELATIVE RESIDUAL 10^{-9}, AND $\lambda_{\min}$ IS COMPUTED BY LANCZOS. ZHOU USES OUR FAST REIMPLEMENTATION.

n/m	HepPh 11,204 / 117,619	AstroPh 17,903 / 196,972	CondMat 21,363 / 91,286	CAIDA 26,475 / 53,381	Enron 33,696 / 180,811
$p=1$	.0639 / 4329.56	.0485 / 4428.97	.0274 / 8874.27	.0656 / 18909.45	.0557 / 17675.78
$p=2$	.0845 / 4332.34	.0841 / 4432.76	.0427 / 8888.70	.0967 / 18915.01	.0759 / 17679.66
$p=3$	.0991 / 4337.35	.0996 / 4437.98	.0506 / 8905.52	.0964 / 18916.07	.0759 / 17679.75
$p=4$	.0991 / 4338.33	.1127 / 4444.54	.0574 / 8927.88	.0990 / 18919.00	.0887 / 17686.85
$p=8$	.1165 / 4358.79	.1193 / 4465.39	.0660 / 8966.20	.1068 / 18927.34	.0887 / 17700.28
Zhou	.1096 / 4343.63	.1127 / 4443.31	.0604 / 8935.30	.1037 / 18921.59	.0888 / 17690.66

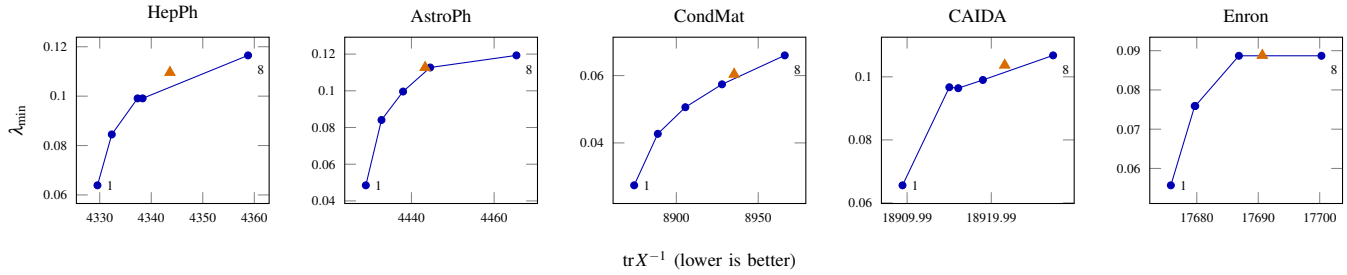


Fig. 4. CFCC/spectral tradeoff on the medium graphs of Table IX. Circles join the sets selected at $p = 1,2,3,4,8$ (labels mark $p = 1$ and $p = 8$); the triangle is Zhou's single set.

TABLE X

PAIRED-PROBE DIFFERENCES $\widehat{\Phi}_1(\cdot) - \widehat{\Phi}_1(\text{GROVE})$ ON THE FINAL SETS OF TABLE II (128 COMMON PROBES), WITH STANDARD ERRORS OF THE DIFFERENCE. POSITIVE VALUES FAVOR GROVE; ROADNET-PA ENTRIES ARE IN THOUSANDS.

Graph	ApproxGreedy	ForestCFCM	SchurCFCM	Zhou
DBLP	$+97.2 \pm 16.7$	$+553.5 \pm 30.3$	$+122.5 \pm 14.2$	$+448 \pm 33$
Amazon	$+266.9 \pm 47.6$	$+1149.6 \pm 74.5$	-164.5 ± 47.6	$+749 \pm 85$
YouTube	$+241.7 \pm 44.3$	$+2637.2 \pm 81.4$	-37.4 ± 44.8	-10.5 ± 39
web-Google	$+404 \pm 79$	$+3247.3 \pm 143.2$	$+302.4 \pm 83.5$	$+661 \pm 98$
Skitter	$+28.8 \pm 22$	$+2001.9 \pm 66.9$	$+434.6 \pm 29.5$	-43.1 ± 19.6
roadNet-PA	$+51.5 \pm 5.9$	-17.0 ± 5.8	-23.8 ± 5.8	$+22.2 \pm 5.6$

reuse of the same projections across adaptive rounds. Nearly-linear solvers and sparse elimination [3], [12] provide the numerical primitives.

VIII. CONCLUSION

GROVE reduces repeated CFCC estimation by updating solution sketches and retaining refreshed moments. Its seeded-gain analysis makes adaptive sampling and numerical conditions explicit. The measurements show both substantial solve savings and limits from maintenance or refresh work, with input-dependent quality tradeoffs. The CM and inverse-power extension supplies a separate choice of spectral objectives.

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